SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:25**

Concept Mapping
Annexure A

Criteria	Concept Identification ***(2.5)***	Linkages ***(2.5)***	Organization ***(2)***	Integration of Literature ***(2)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

M.Pranay
Signature of the student:

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Annexure A

Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

🎯 What your map must show:

1. Core components: Data, Model, User Interface
2. Types of DSS
3. Flow: Data → Analysis → Decision
4. Role in business decision-making
5. Real-world example linkage

Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts

2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

🎯 What your map must show:

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map 5: Query Processing & BI Tools (KNIME)

Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

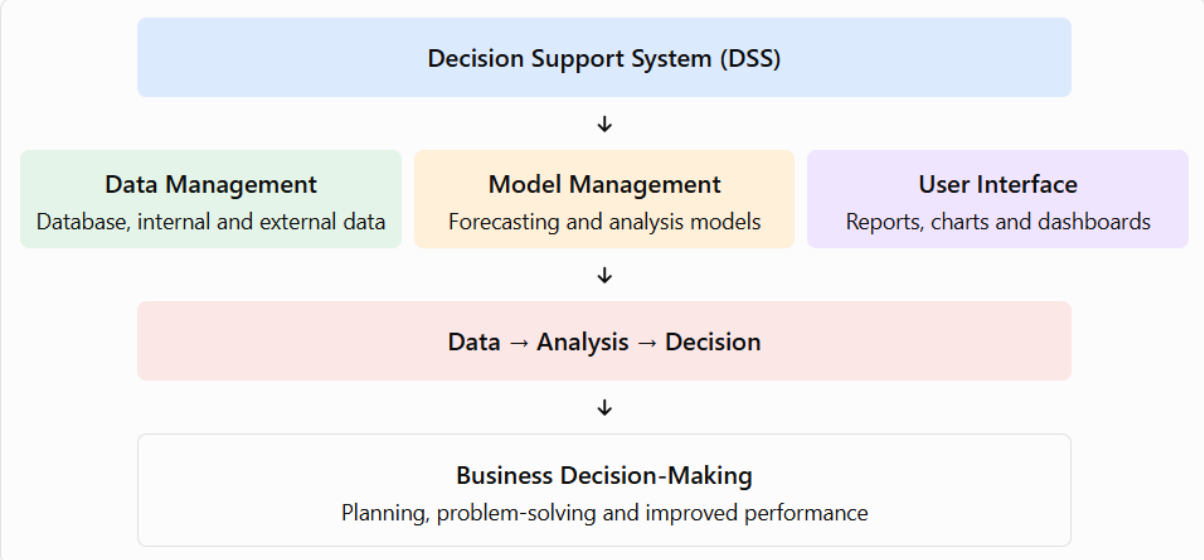
What your map must show:

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integration

ANSWERS

1. Decision Support Systems (DSS)

Definition: A Decision Support System (DSS) is a computer-based system that helps managers make better decisions by analyzing data and using models.



Types of DSS:

- Data-driven DSS
- Model-driven DSS
- Knowledge-driven DSS
- Document-driven DSS
- Communication-driven DSS

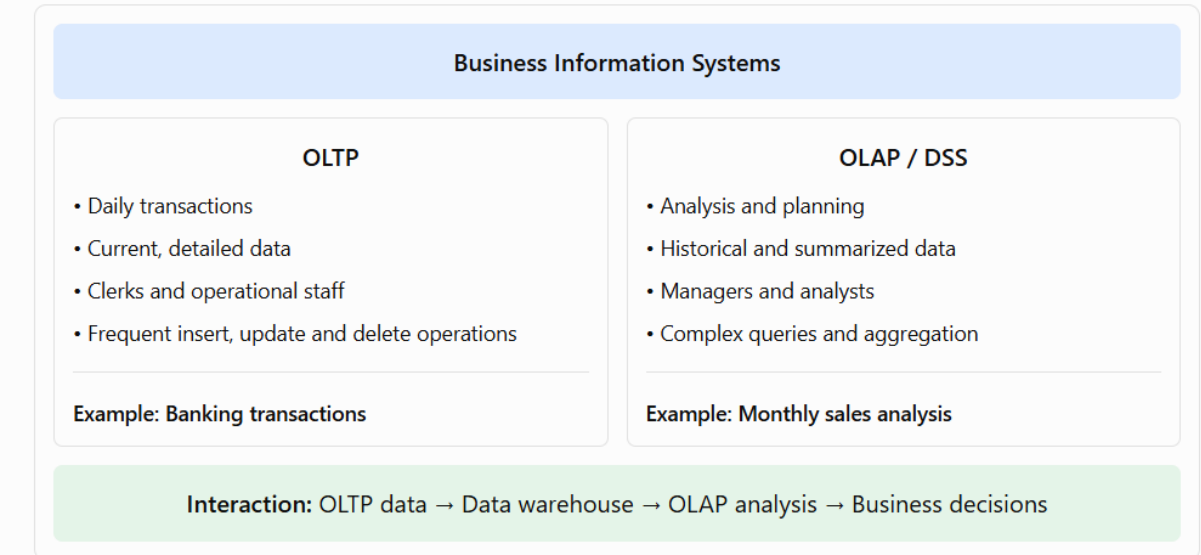
Real-world example: A retail manager uses sales reports and demand forecasts to decide how many laptops to stock for the next month.

Conclusion: DSS helps organizations make informed decisions, reduce uncertainty and improve efficiency.

2. Operational Systems (OLTP) vs DSS (OLAP)

Definitions:

- OLTP (Online Transaction Processing): Processes daily business transactions.
- OLAP (Online Analytical Processing): Analyzes large amounts of data to support decision-making.



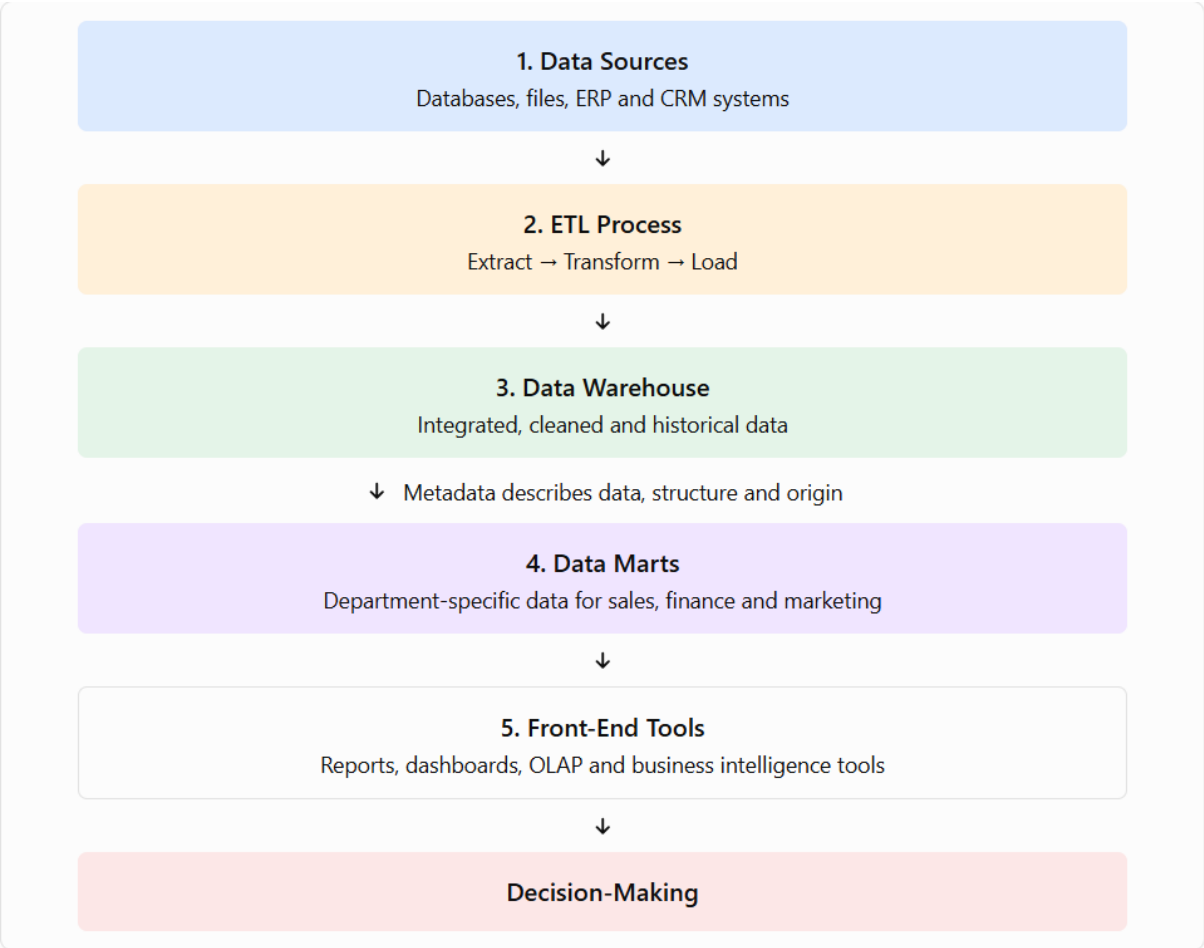
Basis	OLTP	OLAP / DSS
Purpose	Transaction processing	Decision support
Data	Current, detailed	Historical, summarized
Users	Operational staff	Managers and analysts
Processing	Simple, frequent transactions	Complex analytical queries
Example	ATM withdrawal	Bank loan trend analysis

Similarities: Both use data, computer systems and databases to support business operations.

Conclusion: OLTP records daily activities, while OLAP analyzes business data to identify trends and support strategic decisions.

3. Architecture of a Data Warehouse

Definition: A data warehouse is a centralized repository that stores integrated historical data from multiple sources for reporting and analysis.



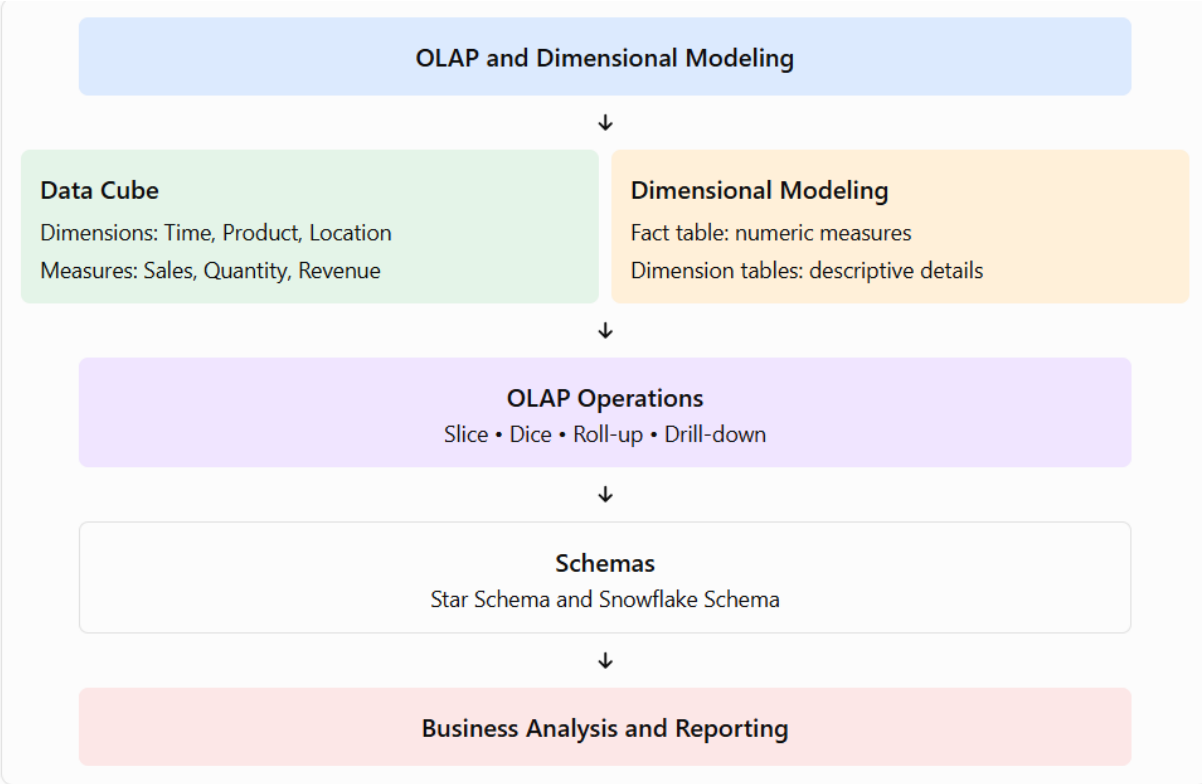
Role of each layer:

1. Data sources: Provide raw data from different systems.
2. ETL: Extracts data, cleans and transforms it, and loads it into the warehouse.
3. Data warehouse: Stores integrated data for long-term analysis.
4. Metadata: Stores information about data definitions, structure and source.
5. Data marts: Provide focused data for individual departments.
6. Front-end tools: Present information through reports, charts and dashboards.

Conclusion: Data warehouse architecture integrates data from different sources and converts it into useful information for organizational decision-making.

4: OLAP, Data Cubes and Dimensional Modeling

Definition: OLAP is a technology used to analyze multidimensional data. Data cubes organize measures across dimensions, while dimensional modeling structures data for efficient analysis.



OLAP operations:

- Slice: Selects one value from a dimension, such as sales in 2026.
- Dice: Selects a subset using multiple dimension conditions.
- Roll-up: Summarizes data, such as daily sales into monthly sales.
- Drill-down: Displays more detailed data, such as monthly sales by day.

Dimensional modeling:

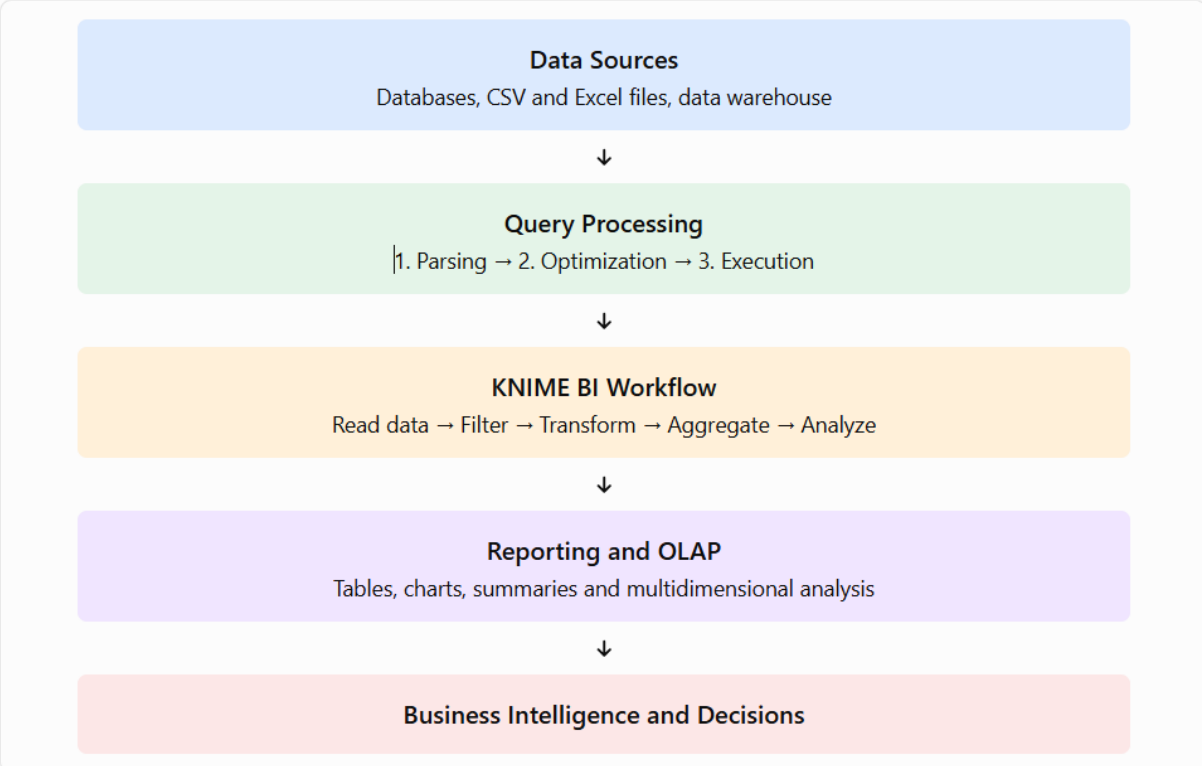
- Fact table: Contains measurable values, such as sales amount and quantity.
- Dimension tables: Contain descriptive information, such as product, date and location.
- Star schema: A central fact table connected directly to dimension tables.
- Snowflake schema: Dimension tables are further normalized into related tables.

Real-world example: An electronics retailer analyzes laptop sales by brand, month and city to identify the best-selling products.

Conclusion: OLAP, data cubes and dimensional modeling help organizations analyze multidimensional data and identify useful business trends.

Q5. Concept Map 5: Query Processing and BI Tools (KNIME)

Definition: Query processing is the process of interpreting and executing database queries. Business Intelligence (BI) tools such as KNIME help users analyze data, create workflows and generate reports.



Query processing stages:

1. Parsing: Checks the query syntax and validates its structure.
2. Optimization: Selects an efficient execution plan.
3. Execution: Runs the query and returns the results.

Role of BI tools:

- Convert raw data into meaningful information.
- Generate reports, charts and dashboards.
- Identify trends and patterns.
- Support planning and decision-making.

KNIME workflow structure:

- Input nodes: Import data from files or databases.
- Processing nodes: Filter, clean, join and aggregate data.
- Analysis nodes: Perform calculations and summarize results.
- Output nodes: Display tables, charts and reports.

OLAP integration: KNIME can prepare and aggregate data for multidimensional analysis, such as comparing laptop sales by brand, region and month. Direct OLAP integration depends on the data source and available extensions.

Conclusion: Query processing and BI tools such as KNIME help transform raw data into useful reports and analytical results for better business decisions.